

Andrew Koch

Bay City, MI | Kochand03@gmail.com | (989) 545-9237

EDUCATION

Michigan State University, College of Engineering
Bachelor of Science, Mechanical Engineering

East Lansing, MI
Expected December 2026

EXPERIENCE

Perrigo Company PLC

Production Engineering Intern

Allegan, MI

May 2025 – August 2025

- Implemented design solutions to improve overall equipment effectiveness on bottle packaging lines
- Designed and implemented adjustable conveyor guides to eliminate bottleneck jams and product loss; supported new induction sealer procurement process to reduce TEP deviations.
- Reduced line flow interruptions and changeover time, helping save over \$100,000 in deviations costs

SKF Aeroengine North America

Digitization Intern

Falconer, NY

May 2024 – August 2024

- Contributed CAD data extraction and reconstruction toward an automated engineering design tool, improving drawing accuracy and reducing discrepancy rates across legacy print library
- Analyzed print data using E-BOM, PLM, and CAD programs to identify and flag design discrepancies, providing actionable improvement points for the design automation program

PROJECTS

Flex Body Modeling & Simulation Best Practices — General Motors Capstone

1st Place, Best Technical Presentation: MSU Engineering Design Day 2026

East Lansing, MI

Jan 2026 – Apr 2026

- Optimized eigenmode selection of suspension components in full-vehicle multibody dynamic simulations. Reduced computational cost while retaining accuracy across ride, handling, and durability loadcases
- Designed components (knuckle, upper/lower control arms, rear axle, differential) in NX, converted to flex bodies in Abaqus including kinematic coupling to nodes and Craig-Bampton substructuring, integrated into the vehicle model in Simpack. Ran simulations and processed data to analyze vehicle and component responses
- Developed and validated two original eigenmode distribution methods: Modal Participation and Component Sensitivity. Achieved up to 8× lower error percentage with equal number of eigenmodes over 150+ simulations.
- Optimized simulations projected to save 6,875 computing hours equating to \$464,000 in annual labor costs per year, directly accelerating vehicle development timeline.

EXTRACURRICULARS

American Society of Mechanical Engineers — MSU Chapter

President (formerly Treasurer)

East Lansing, MI

Jan 2023 – Jan 2025

- Led team in hosting events, technical workshops, and industry talks; launched annual Design Series and 3D Printing Design Competition, brought 10+ companies to MSU for industry talks with student members
- Expanded organizational presence on campus, increasing average event attendance over 75% from prior year

TEDxMSU

Financial Director

East Lansing, MI

Sept 2022 – May 2024

- Led finance team in securing annual conference funding over \$50,000 through sponsorships and fundraisers
- Increasing funding availability by 2.5× since 2021 by pursuing new partnerships with MSU organizations, allowing for larger event operations

RESEARCH

Collatz Conjecture: Quantifying the Chaos

Independent Researcher

April 2024

East Lansing, MI

- Presented mathematics research at the 2024 University Undergraduate Research & Arts Forum
- Created unique visualization to observe problem sequences, derived variables and equations to quantify patterns
- Provided list of multi-variable equations to help simplify the complex and unpredictable nature of the conjecture

SKILLS

Technical: NX (CAD), Shapr3D (CAD), Abaqus (FEA), Simpack (MBD), SimScale (CFD), Python, MATLAB, SAP, Trackwise, Redzone, Lean-Sigma Yellow Belt